

A 25-VERTEX WEAK TENSOR PRODUCT OF α -FREE TREES WITH AN α -VALUATION

ABSTRACT. Anglin, Huczynska and McCourt ask whether the weak tensor product of two graphs that each admit a near α -valuation but no α -valuation must itself admit no α -valuation. The answer is no. Let $S_{3,2}$ be Rosa’s 7-vertex spider, a tree of smallest size with no α -valuation, carrying the near α -valuation printed in their paper. Its weak tensor product with itself, taken with respect to the induced bipartitions, is a 25-vertex 36-edge graph which admits an α -valuation with boundary $\lambda = 27$; the valuation is written out in full, and the check is a pencil computation of 36 differences. The same conclusion holds for the non-isomorphic pair $S_{3,2}, S_{4,2}$, whose product on 32 vertices and 48 edges is exhibited with boundary $\lambda = 36$, so the question is not rescued by reading “two graphs” strictly. Nothing published is contradicted. For *connected* products we record why a positive answer was plausible: the specific product labelling of El-Zanati, Kenig and Vanden Eynden, on which the question rests, is graceful but has no boundary as soon as neither factor labelling is an α -valuation, so it cannot itself witness such a counterexample.

1. INTRODUCTION

All graphs are finite and simple. Let G have q edges. Following Rosa [8], a β -valuation (equivalently, a *graceful labelling*) of G is an injection $b: V(G) \rightarrow \{0, 1, \dots, q\}$ whose multiset of edge differences $\{|b(u) - b(v)| : \{u, v\} \in E(G)\}$ is exactly $\{1, 2, \dots, q\}$. Quoting [1] verbatim, its closing clause set inline rather than displayed:

“An α -valuation b of a graph G with n edges is a β -valuation which satisfies the following extra condition: there exists some x with $0 \leq x \leq n$ such that for each edge $\{u, v\} \in E(G)$: $b(v) \leq x < b(u)$ or $b(u) \leq x < b(v)$.”

We call such an x a *boundary* and write λ for it. Again verbatim from [1], its two-item list set inline:

“A near α -valuation b of a graph G with n edges is a β -valuation which satisfies the following extra condition: there is a partition of $V(G)$ into V_{small} and V_{large} such that for each $v \in V_{\text{small}}$, $b(v) < b(u)$ for all $u \in N(v)$, and for each $v \in V_{\text{large}}$, $b(v) > b(u)$ for all $u \in N(v)$.”

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The notion has three published names: it is Cahit’s 1980 notion [3], the *gracious* labellings of Grannell, Griggs and Holroyd, and the *near α -labelings* of El-Zanati, Kenig and Vanden Eynden [5]; [1] records this coincidence: “This concept was introduced independently in [Cahit], [Grannell] and [El-Zanati], under different names.”

Finally, Snevily’s *weak tensor product* [9], quoted from [1] with its display-math clause:

“The weak tensor product of two bipartite graphs G and H with partitions V_1 and V_2 and W_1 and W_2 respectively is denoted $G \overline{\otimes} H$. It is a graph on the vertices $(V_1 \times W_1) \cup (V_2 \times W_2)$ where vertices (x, y) and (u, v) are adjacent if and only if $\{x, u\} \in E(G)$ and $\{y, v\} \in E(H)$.”

Which bipartition is used is not free. El-Zanati, Kenig and Vanden Eynden [5, Thm. 7], restated in [1], states that if G and H admit near α -valuations γ and δ , then $G \overline{\otimes} H$ *taken with respect to the vertex bipartitions induced by γ and δ* admits a near α -valuation, and [1] adds: “Note that it is essential to use this specific choice of bipartitions.” Throughout, $G \overline{\otimes} H$ means the product on $(V(G)_{\text{small}} \times V(H)_{\text{small}}) \cup (V(G)_{\text{large}} \times V(H)_{\text{large}})$.

The question. The second of the four unnumbered items in the “Further Work” section of [1] reads verbatim and in full, with its cross-reference rendered as the ordinary citation [5, Thm. 7]:

“From Theorem [5, Thm. 7], the weak tensor product of any two graphs with near- α valuations is a graph with a near α -valuation. For two graphs which each possess a near α valuation but no α -valuation, will their weak tensor product have no α -valuation?”

It is a question, not a numbered conjecture: it carries no theorem environment and no number. Read as the universally quantified statement it invites:

(*) *For all bipartite G, H such that G admits a near α -valuation and no α -valuation, and likewise H , the product $G \overline{\otimes} H$ taken with respect to the bipartitions induced by those valuations admits no α -valuation.*

Theorem 1. *(*) is false, and the question quoted above is answered in the negative. Let $S_{3,2}$ be the tree obtained from $K_{1,3}$ by subdividing each edge once, with the near α -valuation γ printed in [1]. Then $S_{3,2}$ admits no α -valuation, while $K = S_{3,2} \overline{\otimes} S_{3,2}$ — a connected bipartite graph on 25 vertices and 36 edges with parts of sizes 16 and 9 — admits the α -valuation displayed in §3, with boundary $\lambda = 27$.*

The same holds for a non-isomorphic pair (§5), so the question is not rescued by insisting that “two graphs” means two distinct graphs. §4 records one reason a positive answer was plausible: when the product is connected,

the particular product labelling of [5], on which the question rests, cannot itself produce a counterexample.

Theorem 7 of [5] (near α in, near α out) and Snevily's theorem [9] (α in, α out) both remain true; what fails is only the hoped-for propagation of α -freeness. See §7 for the limits of what is claimed.

2. THE FACTOR, AND ITS α -FREEDOM

Write $S_{3,2}$ for the spider with centre c , subdivision vertices m_1, m_2, m_3 and leaves l_1, l_2, l_3 , and edges

$$E(S_{3,2}) = \{cm_1, cm_2, cm_3, l_1m_1, l_2m_2, l_3m_3\},$$

so $|V| = 7$ and $q = 6$. Its unique bipartition is $\{c, l_1, l_2, l_3\} \mid \{m_1, m_2, m_3\}$. [1] states that “a tree of smallest size which does not possess an α -valuation is given in [8]; it is the 7-vertex graph $S_{3,2}$ ”, and prints a near α -valuation of it. Read off that figure, the valuation is

$$(1) \quad \gamma(c) = 0, \quad (\gamma(m_i), \gamma(l_i))_{i=1,2,3} = (5, 4), (3, 1), (6, 2).$$

Its six edge differences are 5, 3, 6 (centre edges) and 1, 2, 4 (leaf edges), i.e. exactly $\{1, \dots, 6\}$, so γ is a β -valuation; and with

$$V_{\text{small}} = \{c, l_1, l_2, l_3\} \text{ (labels } 0, 4, 1, 2\text{)}, \quad V_{\text{large}} = \{m_1, m_2, m_3\} \text{ (labels } 5, 3, 6\text{)},$$

every vertex of V_{small} is below and every vertex of V_{large} above all its neighbours, so γ is a near α -valuation. It is not an α -valuation: the two label sets interleave, since $4 \in \gamma(V_{\text{small}})$ exceeds $3 \in \gamma(V_{\text{large}})$.

The following is standard; both later arguments turn on it.

Fact 2. Let T be a connected bipartite graph and let b be a near α -valuation of T with classes $V_{\text{small}}, V_{\text{large}}$.

- (i) b is an α -valuation if and only if $\max b(V_{\text{small}}) < \min b(V_{\text{large}})$, in which case $\lambda = \max b(V_{\text{small}})$ is a boundary.
- (ii) If in addition $|V(T)| = q + 1$, where $q = |E(T)|$, then every α -valuation b' of T with boundary x is a bijection onto $\{0, \dots, q\}$, the two sides of the unique bipartition of T are $\{v : b'(v) \leq x\}$ and $\{v : b'(v) > x\}$, and the first receives exactly the labels $\{0, 1, \dots, x\}$.

Proof. (i) If $\max b(V_{\text{small}}) < \min b(V_{\text{large}})$ then $x = \max b(V_{\text{small}})$ straddles every edge, because every edge of T joins V_{small} to V_{large} . Conversely suppose b has boundary x . The sets $L = \{v : b(v) \leq x\}$ and $U = \{v : b(v) > x\}$ are independent, since every edge straddles x , so $\{L, U\}$ is a bipartition of T ; a connected bipartite graph has only one, so $\{L, U\} = \{V_{\text{small}}, V_{\text{large}}\}$. As every edge runs from V_{small} up to V_{large} we cannot have $L = V_{\text{large}}$, so $L = V_{\text{small}}$ and $\max b(V_{\text{small}}) \leq x < \min b(V_{\text{large}})$.

(ii) Let b' be any α -valuation of T with boundary x . It injects into $\{0, \dots, q\}$ and $|V(T)| = q + 1$, so it is a bijection. Put $L = \{v : b'(v) \leq x\}$ and $U = \{v : b'(v) > x\}$. Every edge straddles x , so both are independent and $\{L, U\}$ is a bipartition of T , hence *the* bipartition. Finally $b'(L) \subseteq \{0, \dots, x\}$ and $|L| = x + 1$ because b' is a bijection, so $b'(L) = \{0, \dots, x\}$. $\square$

Proposition 3. $S_{3,2}$ has no α -valuation.

Proof. Suppose b is one, with boundary x . By Fact 2(ii), b is a bijection onto $\{0, \dots, 6\}$, the low side is one of the two bipartition classes, and it receives an initial segment of labels. Every edge runs low to high, so

$$\sum_{\{u,v\} \in E} |b(u) - b(v)| = \sum_{v \in \text{high}} \deg(v) b(v) - \sum_{v \in \text{low}} \deg(v) b(v) = 1 + \dots + 6 = 21,$$

with $\deg c = 3$, $\deg m_i = 2$, $\deg l_i = 1$.

Case A: $\text{low} = \{c, l_1, l_2, l_3\}$, receiving $\{0, 1, 2, 3\}$, so the m_i receive $\{4, 5, 6\}$. The sum is $2(4+5+6) - (3b(c) + (6 - b(c))) = 24 - 2b(c)$, which is even.

Case B: $\text{low} = \{m_1, m_2, m_3\}$, receiving $\{0, 1, 2\}$, so c and the leaves receive $\{3, 4, 5, 6\}$. The sum is $(3b(c) + (18 - b(c))) - 2(0+1+2) = 2b(c) + 12$, again even.

In both cases the sum is even, but 21 is odd. $\square$

No computation is used; in particular the appeal to [8] is not needed.

3. THE PRODUCT, AND THE α -VALUATION

The graph K , described without reference to the product. K can be built directly. Take a hub h ; a 3×3 grid of vertices Q_{ij} ($1 \leq i, j \leq 3$); three “row” vertices R_1, R_2, R_3 ; three “column” vertices C_1, C_2, C_3 ; and nine pendants P_{ij} . Join h to all nine Q_{ij} ; join R_i to Q_{i1}, Q_{i2}, Q_{i3} ; join C_j to Q_{1j}, Q_{2j}, Q_{3j} ; and join P_{ij} to Q_{ij} . Then $|V(K)| = 1 + 9 + 3 + 3 + 9 = 25$ and $|E(K)| = 9 + 9 + 9 + 9 = 36$. The degree sequence is $9^1 4^9 3^6 1^9$, K is connected, and

$$P = \{h\} \cup \{R_i\} \cup \{C_j\} \cup \{P_{ij}\} \quad (|P| = 16), \quad Q = \{Q_{ij}\} \quad (|Q| = 9)$$

is its unique bipartition.

This is $S_{3,2} \bar{\otimes} S_{3,2}$ with $\gamma = \delta$ as in (1). Indeed, by the definition quoted in §1 the product lives on $(\{c, l_1, l_2, l_3\}^2) \cup (\{m_1, m_2, m_3\}^2)$, i.e. on $16 + 9 = 25$ vertices, and $(x, y) \sim (u, v)$ iff $\{x, u\}$ and $\{y, v\}$ are edges of $S_{3,2}$. Writing xy for (x, y) :

vertex of K	neighbours	degree	rôle
cc	all nine $m_i m_j$	9	h
$l_i c$	$m_i m_1, m_i m_2, m_i m_3$	3	R_i
cl_j	$m_1 m_j, m_2 m_j, m_3 m_j$	3	C_j
$l_i l_j$	$m_i m_j$ only	1	P_{ij}
$m_i m_j$	$cc, cl_j, l_i c, l_i l_j$	4	Q_{ij}

The valuation. Set $\lambda = 27$ and define b on $V(K)$ by the two tables

P -side ($= \{c, l_1, l_2, l_3\}^2$, 16 labels) : $cl_2 = 0$, $l_1l_2 = 1$, $l_1c = 2$, $cl_1 = 3$,
 $l_2l_3 = 4$, $l_2l_2 = 5$, $l_3l_1 = 7$, $l_3c = 8$,
 $l_1l_1 = 9$, $cl_3 = 15$, $l_2c = 17$, $l_3l_3 = 18$,
 $l_3l_2 = 20$, $l_2l_1 = 21$, $l_1l_3 = 23$, $cc = \mathbf{27}$;

Q -side ($= \{m_1, m_2, m_3\}^2$, 9 labels) : $m_1m_1 = \mathbf{28}$, $m_3m_3 = 29$, $m_3m_2 = 30$,
 $m_3m_1 = 31$, $m_2m_2 = 32$, $m_2m_3 = 33$,
 $m_2m_1 = 34$, $m_1m_3 = 35$,
 $m_1m_2 = 36$.

Proof of Theorem 1. Proposition 3 and §2 give the hypotheses: γ is a near α -valuation of $S_{3,2}$ and $S_{3,2}$ has no α -valuation. It remains to check that b above is an α -valuation of K .

Injectivity and range. The 25 values

$$0, 1, 2, 3, 4, 5, 7, 8, 9, 15, 17, 18, 20, 21, 23, 27 \mid 28, 29, \dots, 36$$

are pairwise distinct and lie in $\{0, \dots, 36\} = \{0, \dots, |E(K)|\}$. Exactly $37 - 25 = 12$ labels are unused, namely 6, 10, 11, 12, 13, 14, 16, 19, 22, 24, 25, 26.

The boundary. $\max_P b = 27$ (attained at the hub cc) and $\min_Q b = 28$, while every edge of K joins P to Q . Hence $b(u) \leq 27 < b(v)$ on all 36 edges, so $\lambda = 27$ is a boundary.

Gracefulness. In the rôles of the intrinsic description, $Q_{11} = 28$, $Q_{12} = 36$, $Q_{13} = 35$, $Q_{21} = 34$, $Q_{22} = 32$, $Q_{23} = 33$, $Q_{31} = 31$, $Q_{32} = 30$, $Q_{33} = 29$, and the 36 differences split into four blocks:

$$\begin{aligned} h = cc = 27 &\rightarrow Q_{11}, \dots, Q_{33} : & 1, 9, 8, 7, 5, 6, 4, 3, 2 \\ R_1 = l_1c = 2 &\rightarrow Q_{11}, Q_{12}, Q_{13} : & 26, 34, 33 \\ R_2 = l_2c = 17 &\rightarrow Q_{21}, Q_{22}, Q_{23} : & 17, 15, 16 \\ R_3 = l_3c = 8 &\rightarrow Q_{31}, Q_{32}, Q_{33} : & 23, 22, 21 \\ C_1 = cl_1 = 3 &\rightarrow Q_{11}, Q_{21}, Q_{31} : & 25, 31, 28 \\ C_2 = cl_2 = 0 &\rightarrow Q_{12}, Q_{22}, Q_{32} : & 36, 32, 30 \\ C_3 = cl_3 = 15 &\rightarrow Q_{13}, Q_{23}, Q_{33} : & 20, 18, 14 \\ P_{ij} &\rightarrow Q_{ij} : & 19, 35, 12, \quad 13, 27, 29, \quad 24, 10, 11 \end{aligned}$$

(the last line lists P_{11}, P_{12}, P_{13} , then P_{21}, P_{22}, P_{23} , then P_{31}, P_{32}, P_{33} , whose labels are 9, 1, 23, then 21, 5, 4, then 7, 20, 18). Sorting the union and tagging each value with the block it came from:

$$\begin{array}{cccccccccccccccccccccccc} 1_h & 2_h & 3_h & 4_h & 5_h & 6_h & 7_h & 8_h & 9_h & 10_P & 11_P & 12_P & 13_P & 14_C & 15_R & 16_R & 17_R & 18_C & 19_P & 20_C & 21_R & 22_R & 23_R & 24_P \\ 25_C & 26_R & 27_P & 28_C & 29_P & 30_C & 31_C & 32_C & 33_R & 34_R & 35_P & 36_C \end{array}$$

which is $1, 2, \dots, 36$, each exactly once. So b is a β -valuation, and with the boundary above it is an α -valuation. $\square$

Remark. Two independent consistency checks on the printed numbers. First, summing $b(\text{high end}) - b(\text{low end})$ over all edges gives $\sum_{v \in Q} \deg(v)b(v) - \sum_{v \in P} \deg(v)b(v) = 666 = 1 + 2 + \dots + 36$. Second, the edge carrying difference 36 is $cl_2 = 0 \sim m_1m_2 = 36$, and those two vertices are adjacent, since $\{c, m_1\}$ and $\{l_2, m_2\}$ are edges of $S_{3,2}$.

Corollary 4. *α -freeness is not preserved by the weak tensor product, even for trees as factors and even when the product is taken with respect to the bipartitions induced by near α -valuations of the factors.*

4. THE PRODUCT LABELLING σ

When the product is connected, the labelling of $G \bar{\otimes} H$ on which the question rests cannot itself witness a counterexample. Let $A = V(G)_{\text{small}}$, $B = V(G)_{\text{large}}$, $m = |E(G)|$, $g_A = \max \gamma(A)$, $g_B = \min \gamma(B)$, and let C, D, n, h_C, h_D be the corresponding data for H and δ . The labelling σ of $G \bar{\otimes} H$ used by [5] and reproduced in [1] is

$$\begin{aligned} \sigma(a, c) &= m \delta(c) + \gamma(a) & (a \in A, c \in C), \\ \sigma(b, d) &= m (\delta(d) - 1) + \gamma(b) & (b \in B, d \in D), \end{aligned}$$

and is a β -valuation of $G \bar{\otimes} H$, with edge differences $m(f - 1) + e$ for $e \in \{1, \dots, m\}$, $f \in \{1, \dots, n\}$.

Lemma 5. *Suppose $G \bar{\otimes} H$ is connected. Then σ is an α -valuation of $G \bar{\otimes} H$ if and only if*

$$m(h_C - h_D + 1) < g_B - g_A.$$

Proof. Every edge of $G \bar{\otimes} H$ runs from $A \times C$ to $B \times D$, and σ is a near α -valuation with those classes [5, Thm. 7]. By Fact 2(i), σ is an α -valuation iff $\max \sigma(A \times C) < \min \sigma(B \times D)$. Now $\max \sigma(A \times C) = m h_C + g_A$ and $\min \sigma(B \times D) = m(h_D - 1) + g_B$, and rearranging gives the claim. $\square$

Corollary 6. *Suppose $G \bar{\otimes} H$ is connected. If neither γ nor δ is an α -valuation, then σ is not an α -valuation of $G \bar{\otimes} H$ — for every such choice of γ and δ and in either coordinate order. Conversely, if both γ and δ are α -valuations then σ is one, which is Snevily's theorem [9] for $\bar{\otimes}$.*

Proof. Labels are distinct, so by Fact 2(i) exactly one of $g_A < g_B$, $g_A > g_B$ holds, and γ is α iff the first does; likewise for δ . If neither is α then $g_B - g_A \leq -1$ and $h_C - h_D \geq 1$, so $m(h_C - h_D + 1) \geq 2m \geq 2 > -1 \geq g_B - g_A$ and Lemma 5 fails. If both are α then $g_B - g_A \geq 1$ and $h_C - h_D \leq -1$, so $m(h_C - h_D + 1) \leq 0 < 1 \leq g_B - g_A$. $\square$

Connectedness of $G \bar{\otimes} H$ is used in both statements, through the uniqueness of the bipartition in Fact 2(i); we do not settle what σ does when $G \bar{\otimes} H$ is disconnected. Both products exhibited in this note are connected.

For our K , $m = 6$, $h_C = g_A = 4$ and $h_D = g_B = 3$, so the criterion reads $12 < -1$: σ gives labels with $\max_P \sigma = 28 > 15 = \min_Q \sigma$, graceful but with no boundary. The witness b of §3 is therefore not σ ; it is also not *separable*,

i.e. not of the form $\varphi(x, y) = p(x) + q(y)$, since $b(cl_1) - b(cl_2) = 3$ while $b(l_1l_1) - b(l_1l_2) = 8$, which separability would force to be equal. We do not claim that a counterexample must be non-separable.

5. TWO DISTINCT GRAPHS

Theorem 7 of [5], on which the bullet explicitly rests, has no distinctness or non-isomorphism hypothesis, so Theorem 1 already answers it. The strict reading of “two graphs” does not rescue it either.

Let $S_{4,2}$ be the comet with centre c , subdivision vertices $a_0, \dots, a_3$ and leaves $b_0, \dots, b_3$, edges ca_j and b_ja_j ; so $|V| = 9$, $q = 8$, and the unique bipartition is $\{c, b_0, \dots, b_3\} \mid \{a_0, \dots, a_3\}$. The labelling

$$(2) \quad \delta(c) = 0, \quad (\delta(a_j), \delta(b_j))_{j=0,1,2,3} = (8, 4), (7, 2), (6, 5), (3, 1)$$

is a bijection onto $\{0, \dots, 8\}$ with edge differences 8, 7, 6, 3 from the centre edges and 4, 5, 1, 2 from the leaf edges, i.e. exactly $\{1, \dots, 8\}$. It is a near α -valuation, with V_{small} the set of c and the four leaves; and it is not an α -valuation, since $5 = \delta(b_2)$ exceeds $3 = \delta(a_3)$.

Proposition 7. $S_{4,2}$ has no α -valuation.

Proof. As in Proposition 3, an α -valuation b is a bijection onto $\{0, \dots, 8\}$, the low side is a bipartition class receiving an initial segment, and the edge differences sum to $1 + \dots + 8 = 36$. Here parity alone does not decide, so we push one step further.

Case A: $low = \{c, b_0, \dots, b_3\}$, receiving $\{0, 1, 2, 3, 4\}$; the a_j receive $\{5, 6, 7, 8\}$. The sum is $2(5+6+7+8) - (4b(c) + (10 - b(c))) = 42 - 3b(c)$, so $b(c) = 2$ is forced. Then the centre edges contribute the differences $\{5-2, 6-2, 7-2, 8-2\} = \{3, 4, 5, 6\}$, so the four leaf edges must contribute $\{1, 2, 7, 8\}$. The leaves receive $\{0, 1, 3, 4\}$ and each is matched with a distinct a_j . The only pair in $\{5, 6, 7, 8\} \times \{0, 1, 3, 4\}$ with difference 1 is $(5, 4)$, and the only ones with difference 2 are $(5, 3)$ and $(6, 4)$; using $(5, 4)$ consumes both 5 and 4, so differences 1 and 2 cannot both occur.

Case B: $low = \{a_0, \dots, a_3\}$, receiving $\{0, 1, 2, 3\}$; then c and the leaves receive $\{4, \dots, 8\}$. The sum is $(4b(c) + (30 - b(c))) - 2(0+1+2+3) = 3b(c) + 18$, so $b(c) = 6$ is forced. The centre edges contribute $\{6, 5, 4, 3\}$, so again the leaf edges must contribute $\{1, 2, 7, 8\}$. The leaves receive $\{4, 5, 7, 8\}$; the only pair with difference 1 is $(4, 3)$ and the only ones with difference 2 are $(4, 2)$ and $(5, 3)$, and $(4, 3)$ consumes both 4 and 3. Same contradiction. $\square$

Theorem 8. $(*)$ fails on a non-isomorphic pair. With γ as in (1) on $G = S_{3,2}$ (relabel $m_i \mapsto a_{i-1}$, $l_i \mapsto b_{i-1}$) and δ as in (2) on $H = S_{4,2}$, the product $L = G \otimes H$ is a connected bipartite graph on 32 vertices and 48 edges

with parts of sizes 20 and 12, and

$$\begin{aligned} P \text{ (low, 20)} : \quad & b_0b_1 = 0, \quad b_0c = 1, \quad cb_3 = 2, \quad cb_0 = 3, \quad cb_2 = 4, \\ & b_2b_0 = 5, \quad b_2c = 9, \quad b_2b_2 = 10, \quad b_1c = 17, \quad b_1b_3 = 18, \\ & b_1b_1 = 19, \quad b_1b_2 = 20, \quad b_0b_0 = 22, \quad b_1b_0 = 23, \quad b_2b_1 = 24, \\ & cb_1 = 25, \quad b_2b_3 = 26, \quad b_0b_2 = 27, \quad b_0b_3 = 28, \quad cc = \mathbf{36}; \end{aligned}$$

$$\begin{aligned} Q \text{ (high, 12)} : \quad & a_0a_0 = \mathbf{37}, \quad a_1a_1 = 38, \quad a_2a_2 = 39, \quad a_2a_3 = 40, \\ & a_2a_1 = 41, \quad a_2a_0 = 42, \quad a_1a_3 = 43, \quad a_1a_2 = 44, \\ & a_1a_0 = 45, \quad a_0a_3 = 46, \quad a_0a_2 = 47, \quad a_0a_1 = 48 \end{aligned}$$

is an α -valuation of L with boundary $\lambda = 36$.

Proof. Here $(x, y) \sim (u, v)$ iff $\{x, u\} \in E(G)$ and $\{y, v\} \in E(H)$, on

$$(\{c, b_0, b_1, b_2\} \times \{c, b_0, \dots, b_3\}) \cup (\{a_0, a_1, a_2\} \times \{a_0, \dots, a_3\}),$$

i.e. on $20 + 12 = 32$ vertices; the degrees are 12 at cc , 3 at each cb_j , 4 at each b_ic , 1 at each b_ib_j , so $|E(L)| = 12 + 4 \cdot 3 + 3 \cdot 4 + 12 = 48$, and L is connected. The 32 labels above are distinct and lie in $\{0, \dots, 48\}$; $\max_P = 36 < 37 = \min_Q$ and every edge runs $P \rightarrow Q$, so $\lambda = 36$ straddles all 48 edges. That the 48 differences are exactly $\{1, \dots, 48\}$ is a finite check on the printed labels; as a consistency test, $\sum_{v \in Q} \deg(v)b(v) - \sum_{v \in P} \deg(v)b(v) = 1176 = 1 + 2 + \dots + 48$. Propositions 3 and 7 supply the hypotheses. $\square$

6. RELATION TO KNOWN RESULTS

Every weak-tensor-product theorem recorded in Gallian's survey [6] requires the factors to *have* an (α or near- α) labelling: Snevily [9], El-Zanati–Kenig–Vanden Eynden [5, Thm. 7], and the generalisation of López and Muntaner-Batle [7]. We found no statement in either direction there about $\boxtimes$ applied to α -free factors, which is exactly the gap the question of [1] occupies.

The nearest results should be recorded, since the phenomenon — an α -free tree sitting inside an α -labelled multi-copy host — is *not* new. Ahmed and Snevily [2] prove that for every comet T either T has an α -labelling or there is a graph H_T with an α -labelling that decomposes into two edge-disjoint copies of T ; the factors used here, $S_{3,2}$ and $S_{4,2}$, are comets, so their result already applies to them. Their host is not a weak tensor product, so it does not bear on (*). El-Zanati, Fu and Shiue [4] prove Snevily's conjecture that every bipartite graph has a finite α -labelling number, i.e. that some number of edge-disjoint copies of it tile a graph carrying an α -labelling; again the host is not a weak tensor product, and nothing there decides (*). And López and Muntaner-Batle [7], which [1] does not cite, enlarges the classes of graphs admitting α -, near- α - and bigraceful labellings — but its α -conclusion theorem requires α -labellings on both factors, and its near- α theorem concludes only near α . Anglin, Huczynska and McCourt note a

further adjacency: “However as shown in [Barrientos], graphs which contain this as a subgraph may still have α -valuations.”

What settles the question of [1] is that the product construction the question itself names is such a host. We do not claim that $\bar{\otimes}$ never yields α -free graphs from α -free factors: the two products above are single instances.

7. SCOPE

- What is refuted is $(*)$, the universally quantified reading of an interrogative bullet. [1] poses no conjecture there, and we do not claim it does. Nothing published is contradicted; Theorem 7 of [5] and Snevily’s theorem both stand, and Lemma 5 recovers the latter for connected products.
- No general theorem in the opposite direction is proved: we do *not* claim that every such product has an α -valuation.
- Nothing is claimed about α -free near- α trees beyond the two factors used here, nor about non-tree α -free bipartite factors, nor about minimality: we do not claim K is the smallest counterexample.
- One residual is definitional rather than arithmetical: that “near α -valuation” and “weak tensor product” are read as [1] intends. Every quotation above is verbatim from the L^AT_EX e-print of [1], with no word added, dropped or altered; the only editing is of formatting, in that the source’s displayed formulae and its two-item list in the near α -valuation definition are set inline here, and its citation and cross-reference macros are rendered as bracketed references. The reading was pinned by reproducing that paper’s own printed valuation (1) and its stated properties, but a definitional misreading is not something arithmetic can exclude.
- The accompanying program `verify.py`, with its recorded output `verify.output.txt`, re-checks the objects printed above and the finite claims made about them. It does not search for further counterexamples, does not address minimality, and establishes nothing about factors on more than 9 vertices.

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